

Lista de Exercícios 5

Macroeconomia 1 – MPE Insper

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1. (5 pontos) Based on Kurlat (2020, Cap. 9)

Consider a two-period economy with a representative household, a representative firm, and competitive markets for goods, labor and capital. There is no government and no investment technology: the capital stock is fixed at $\bar{K}$ in both periods and does not depreciate. The household owns $\bar{K}$ and rents it to the firm each period at the rental rate r_t^K .

In each period the household has one unit of time, which it splits between leisure l_t and work L_t . Its preferences are

$$u(c_1) + v(l_1) + \beta [u(c_2) + v(l_2)]$$
$$\text{with } u(c) = \frac{c^{1-\sigma}}{1-\sigma}, \quad v(l) = \psi \ln l$$

where $\beta \in (0, 1)$, $\sigma > 0$ and $\psi > 0$. The household can borrow and lend at the interest rate r , and receives the profits Π of the firm.

The firm produces in each period with the technology

$$F(K_t, L_t) = A_t K_t^\alpha L_t^{1-\alpha}$$

where $\alpha \in (0, 1)$ and $A_t > 0$ is total factor productivity.

- Write down the household's problem with its present-value budget constraint, and the firm's problem in each period. Explain why $\Pi = 0$ in equilibrium.
- Derive the household's first order conditions and the firm's first order conditions. Identify which one describes the trade-off between consumption and leisure and which one describes the trade-off between present and future consumption.
- Define a competitive equilibrium for this economy, listing the allocation, the prices, and the conditions they must satisfy. Include the market clearing conditions for goods, labor and capital in each period.
- Combine the conditions in item (b) with market clearing to obtain the equation that determines equilibrium labor L_t , and write it in the form "marginal rate of substitution = marginal rate of transformation". Then use the Euler equation to show how the interest rate r is determined.
- Consider an increase in A_t . Show that

$$\frac{d \ln L_t}{d \ln A_t} = \frac{1 - \sigma}{\alpha + (1 - \alpha)\sigma + \frac{L_t}{1-L_t}}$$

and use it to sign the effect on w_t , r_t^K and c_t . For which values of σ does higher productivity make the household work more? Interpret in terms of income and substitution effects.

2. (5 pontos) Based on Kurlat (2020, Cap. 9)

Consider an infinite-period economy with a representative household, a representative firm, and competitive markets. Labor supply is fixed at $L_t = 1$ for every period (households don't value

leisure and work all the available hours). The household discounts the future at rate $\beta \in (0, 1)$ and has preferences

$$\mathcal{U} = \sum_{t=0}^{\infty} \beta^t u(c_t), \quad u(c) = \frac{c^{1-\sigma}}{1-\sigma}, \quad \sigma > 0$$

The firm produces with

$$F(K_t, L_t) = AK_t^\alpha L_t^{1-\alpha}, \quad \alpha \in (0, 1)$$

Capital depreciates at rate $\delta \in (0, 1)$ and the initial capital stock K_0 is given.

- a) Write down the household's problem and its budget constraint, and derive the Euler equation. Explain the intuition of this equation and what makes consumption grow over time. What role does σ play?
- b) Solve the firm's problem and obtain w_t and r_t^K .
- c) Capital has to be owned by someone. Compare two arrangements: (i) the household owns the capital stock, rents it to the firm each period at r_t^K , and does the investment itself; (ii) a separate investment firm builds and holds the capital and rents it to the productive firm. Show that both arrangements imply the same no-arbitrage condition

$$1 + r_{t+1} = r_{t+1}^K + 1 - \delta$$

and explain why the equilibrium allocation therefore does not depend on who owns the capital.

- d) Using the goods market clearing condition and the capital accumulation equation, write the resource constraint of this economy. Combine it with items (a)–(c) to show that the steady state capital stock satisfies

$$F_K(K_{ss}, 1) - \delta = \frac{1}{\beta} - 1$$

Solve for K_{ss} and c_{ss} in closed form. What is the steady state interest rate, and why does it depend only on β ?

- e) (*Bonus.*) The Golden Rule capital stock K_{gr} is the one that maximizes steady state consumption. Derive it and show that $K_{ss} < K_{gr}$. Since the First Welfare Theorem tells us the competitive equilibrium is optimal, why does this economy not accumulate enough capital to reach the Golden Rule? What happens as $\beta \rightarrow 1$?